

PRACTILE NO :-3

Name:- Gayatri Hemant Pawar

Roll No: 272

Batch:-S3

Code:-

```
#include <iostream>

using namespace std;

struct Node {
    int data;
    Node *next;
};

Node *front = NULL, *rear = NULL;

// isEmpty
bool isEmpty() {
    return front == NULL;
}

// isFull
bool isFull() {
    Node *p = new(nothrow) Node;
    if (p == NULL) return true;
    delete p;
    return false;
}

// Enqueue
void enqueue(int x) {
    Node *p = new Node;
```

```
p->data = x;
```

```
if (isEmpty())
```

```
    front = rear = p;
```

```
else {
```

```
    rear->next = p;
```

```
    rear = p;
```

```
}
```

```
rear->next = front;
```

```
}
```

```
// Dequeue
```

```
void dequeue() {
```

```
    if (isEmpty()) {
```

```
        cout << "Empty\n";
```

```
        return;
```

```
    }
```

```
    Node *p = front;
```

```
    if (front == rear)
```

```
        front = rear = NULL;
```

```
    else {
```

```
        front = front->next;
```

```
        rear->next = front;
```

```
    }
```

```
    delete p;
```

```
}
```

```
// Display
```

```
void display() {  
    if (isEmpty()) {  
        cout << "Empty\n";  
        return;  
    }  
  
    Node *p = front;  
  
    do {  
        cout << p->data << " ";  
        p = p->next;  
    } while (p != front);  
  
    cout << endl;  
}
```

```
int main() {  
    enqueue(10);  
    enqueue(20);  
    enqueue(30);  
  
    display();  
  
    dequeue();  
    display();  
  
    return 0;  
}
```

Output:-

10 20 30

20 30